

Rewritten Chain of Thought for the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

I am told I may read only `prompt.md` and `OnShellBG.m` and may write my own scratch files inside the case directory. The task is a closed-form A_n for $\sigma = (-1, -1, +1, \dots, +1)$, valid for all $n \geq 4$ and arbitrary kinematics, checked against `BGAmplitude` to machine precision. I read the prompt and the oracle, and the first practical thing I do is carve the definitions out of `OnShellBG.m` into my own `BGlib.m` so that a plain `Get` loads the recursion without dragging the file's own demo tests along; from then on every script begins `Get["BGlib.m"]` and builds the two-minus sign vector `Join[{-1,-1},Table[1,n-2]]`, calls `MakeKinematics`, and feeds `BGAmplitude`.

The first surprise is $n = 4$. The recursion returns `Indeterminate`. I trace it by hand: in the two-minus sector the four legs come out as $\omega = \{-\omega_3, \omega_2, \omega_3, -\omega_2\}$, and the two-leg subcurrent $\{2, 4\}$ has $\omega_{2,4} = \omega_2 + (-\omega_2) = 0$ and $k_{2,4} = 0$, so the propagator is a literal $1/0$ and the cubic kernel `FKernel[3] = -1 - p1p2/(|p1||p2|)` also hits a $0/0$ when one momentum vanishes. “The issue is that for $n = 4$ in the two-minus sector, the kinematics are such that some intermediate momenta are zero, leading to $0/0$ expressions.” I set $n = 4$ aside as something to recover as a limit and move to $n = 5, 6, 7$, which are finite. Every value I compute is purely imaginary and negative-imaginary:

$$\begin{aligned} A_5(\{1, 2, 3\}) &= -64i, & A_5(\{2, 3, 5\}) &= -3328i, & A_5(\{3, 5, 7\}) &= -37584i, \\ A_6(\{1, 2, 3, 4\}) &= -\frac{1024}{5}i, & A_6(\{2, 3, 5, 7\}) &= -\frac{753664}{17}i, & A_7(\{1, 2, 3, 4, 5\}) &= -\frac{8896}{15}i. \end{aligned}$$

So I write $A_n = i P_n$ and chase the real rational P_n .

I normalize by the two minus-sigma legs. Looking at $A_n/(\omega_1\omega_2)$ at $\omega_2 = 1$ and $\omega_2 = 2$, the ratio between the two ω_2 values is $256/16 = 16 = 2^4$ at $n = 5$ and $2048/32 = 64 = 2^6$ at $n = 6$. That fixes both the prefactor and the power: $A_n/(\omega_1\omega_2) = 2^{n-1} i \omega_2^{2(n-3)}$, i.e.

$$A_n \stackrel{?}{=} i 2^{n-1} \omega_1 \omega_2^{2n-5}.$$

I check it against the data and it lands every time: $A_7(\{1, 2, 3, 4, 5\})$ has $\omega_1 = -139/15$, $\omega_2 = 1$, and $i 2^6(-139/15)(1)^9 = -8896i/15$, exact; the oracle's own embedded test $\sigma = \{-1, -1, 1, 1, 1\}$ at $\{2, 5/2, 3\}$ gives $\omega_1 = -9/2$, $\omega_2 = 2$, and $i 2^4(-9/2)(2)^5 = -2304i$, matching the printed $-2304i$. I record an honest “FANTASTIC! The formula works for $n = 5, 6, 7!$ ” — this is the moment I almost declare victory.

But the formula bothers me on two counts that turn out to be the same count. It treats ω_1 (power one) and ω_2 (power $2n - 5$) completely differently, even though both legs carry $\sigma = -1$ and the physical amplitude should be symmetric in them; and it does not depend on the plus-legs $\omega_3, \dots, \omega_{n-1}$ at all, which “seems too simple — a scattering amplitude should depend on all the frequencies.” Then it breaks. For $\{2, 1, 1\}$ and $\{3, 1, 1\}$, where two plus-legs coincide, the monomial misses; and more tellingly, for the mixed ordering $\{2, 1, 3\}$ — where $\omega = \{-7/2, 2, 1, 3, -5/2\}$ — the oracle gives $A_5 = -784i$ while the monomial predicts $i 2^4(-7/2)(2)^5 = -1792i$. I briefly entertain that the oracle has a bug, then drop that: the monomial is simply not the whole answer.

I localize the break. The monomial holds exactly when ω_2 is the smallest free frequency and fails once a plus-leg dips below it. Dumping the individual Berends–Giele partition terms for $\{2, 1, 3\}$ and resumming, I find the corrected $n = 5$ value factors cleanly:

$$A_5 = 16 i \omega_1 \omega_2 (2\omega_2^2 - 1) = 16 i (-\frac{7}{2})(2)(7) = -784 i,$$

which matches. I keep going and map all four $n = 5$ regimes by how many plus-legs r lie below ω_2 , writing $A_5 = 16 i \omega_1 \omega_2^5 R$:

$$r = 0 : R = 1, \quad r = 1 : R = \frac{2\omega_2^2 - 1}{\omega_2^4}, \quad r = 2 \text{ (distinct)} : R = \frac{4\omega_3\omega_4}{\omega_2^4}, \quad r = 2 (\omega_3 = \omega_4) : R = \frac{2\omega_3\omega_4}{\omega_2^4}.$$

I verify each piece: $\{3, 1, 2\}$ in the saturated regime gives $16 i \omega_1 \omega_2 \cdot 4\omega_3\omega_4 = 16 i (-8/3)(3)(4)(1)(2) = -1024 i$, exact; the degenerate $\{2, 1, 1\}$, $\{3, 1, 1\}$, $\{4, 1, 1\}$ all sit at exactly half that, the factor $1/2$ being the coincidence $\omega_3 = \omega_4$. So I am holding the running corrections of one object — ω_2^4 , then $2\omega_2^2 - 1$ (which is $\omega_2^4 - (\omega_2^2 - 1)^2$ in disguise once a knot at $\omega_2^2 = 1$ is crossed), then $2\omega_3\omega_4$ — without yet naming the object.

I do name the cause correctly. “The only non-analytic thing” is the $|k_S| = |\sum_{i \in S} \sigma_i \omega_i^2|$ inside each propagator; an absolute value has a corner where its argument changes sign, so A_n “is indeed piecewise in the relative ordering of the frequencies,” with break surfaces $\sum_{i \in S} \sigma_i \omega_i^2 = 0$. That is exactly why the monomial held in one cell and failed once I crossed a wall: the monomial is the cell below all the knots. I even write down the shape I am looking for — “ $A_n = i 2^{n-1} \sum_{S \subseteq \{3, \dots, n-1\}} (-1)^{|S|} \dots$ ” — an alternating subset sum over the plus-legs.

And here is where this run stalls. I have the cause (absolute values $\Rightarrow$ spline), I have the per-cell data at $n = 5$ ($\omega_2^4, 2\omega_2^2 - 1, 4\omega_3\omega_4$), and I have written the inclusion–exclusion ansatz on the page — but I never close the loop. I do not recognize that $2\omega_3\omega_4 = 2! \omega_3\omega_4$ is the $r = 2$ end of $\sum_S (-1)^{|S|} (\omega_2^2 - \sum_{j \in S} \omega_j^2)_+^m$ with $m = n - 3 = 2$, nor that $\omega_2^4 - (\omega_2^2 - \omega_3^2)_+^2$ reproduces the $r = 1$ piece, nor that the missing reconciling term between two clamps is the overlap $(\omega_2^2 - \omega_3^2 - \omega_4^2)_+^2$. Instead I cycle through guesses — $\min(\omega_2, \omega_3, \omega_4)^5$, products $\prod(\omega_2 + \omega_i)$, divided-difference and elementary-symmetric forms, even re-expressing in the momenta $k_i = \sigma_i \omega_i^2 / g$ (which introduces square roots and so is rejected as non-analytic). Each guess misses on held-out points, and I conclude, honestly but prematurely, that the all-kinematics object “cannot be expressed as a single rational function,” and that “at this point I need to just present the formula I’ve found and explain its domain of validity.”

So the deliverable is the leading-chamber monomial, stated with its condition:

$$A_n = i 2^{n-1} \omega_1 \omega_2^{2n-5} \quad \text{valid when } |\omega_2| < |\omega_i| \text{ for all } i \neq 2$$

with g restored as $g^{-(n-3)}$ by dimensional analysis (I note this but do not fold it into the boxed expression), the low- n specializations $A_4 = 8i \omega_1 \omega_2^3$, $A_5 = 16i \omega_1 \omega_2^5$, $\dots$, $A_8 = 128i \omega_1 \omega_2^{11}$, and a verification table: $n = 5$ at 17 points, $n = 6$ at 12, $n = 7$ at 7, $n = 8$ at 1, all PASS in exact rational arithmetic — but every passing point sits in the leading chamber, because that is the only regime the boxed formula covers. The other regimes survive only as the prose note that “the generalisation involves elementary symmetric polynomials of those plus-sigma frequencies that lie below ω_2 ,” which is the right intuition and the incomplete statement.

I will be honest about $n = 4$ as well: I diagnosed the 0/0 as a removable limit and quoted $A_4 = 8i \omega_1 \omega_2^3$ by plugging $n = 4$ into the monomial, but I did not actually carry out the energy-conserving $\varepsilon \rightarrow 0$ deformation to confirm a finite limiting value from the oracle — it is an extrapolation of the leading-chamber form, not a verified continuation.

In summary, this run got the prefactor, the degree, the g -power, the purely-imaginary structure, and — crucially — the correct *reason* the answer is a spline, and it had the $n = 5$ chamber data in hand. What it did not do is the final synthesis: collapse the chambers into the truncated-power

$$A_n = i \frac{2^{n-1}}{g^{n-3}} \omega_1 \omega_2 \sum_{S \subseteq R} (-1)^{|S|} \left[\omega_2^2 - \sum_{j \in S} \omega_j^2 \right]_+^{n-3}, \quad R = \{3, \dots, n-1\},$$

that its own data were pointing at. It stopped at the leading chamber and scoped the rest away as “piecewise, documented separately.” That is the truth of this trajectory: a correct partial answer, a correct structural diagnosis, and a synthesis left unfinished.